Hazel Chen

Rising 3rd Year ME and EE Student at University of Virginia

Project Portfolio

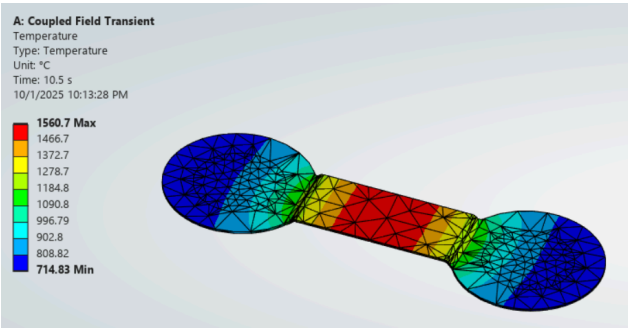
https://www.linkedin.com/in/hazelchens/
https://github.com/Hazel-H-Chen

Model of fusible links for use in a cylindrical cell battery pack

Objective

Fuses in a battery pack isolate faulty cells and prevent dangerous current surges. In cylindrical-cell packs, spot-welded nickel strips often serve double duty as both cell interconnects and fusible links, so their geometry has to be tuned to fuse at a target fault current without failing during normal operation.

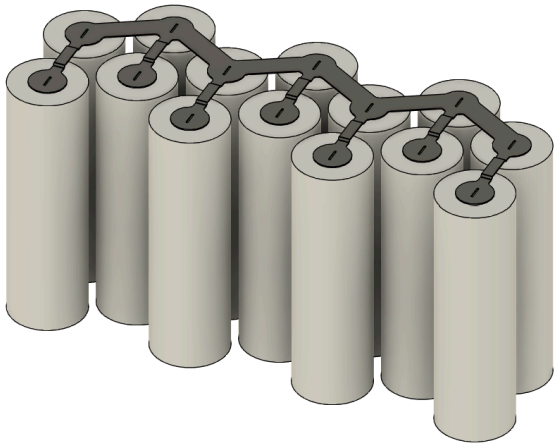
This project set out to determine what nickel strip geometry fuses reliably at a specified current. To evaluate candidate geometries, I modeled cell temperature over time in Ansys for a given current and strip design, using Joule heating to predict whether and when the link reaches its fusing point.



Temperature at 10.5s, 50A fault current

Future Plans

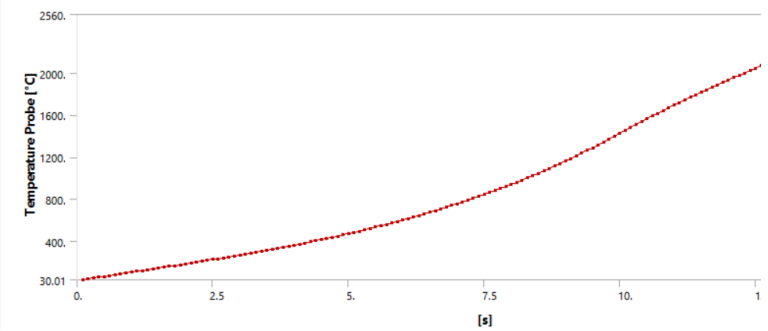
The current model treats the nickel strip in isolation, which makes it a conservative baseline. Planned extensions include modeling conductive heat transfer into the connected cells, which act as heat sinks that would raise the true fusing current; simulating dynamic current profiles rather than a fixed current, to reflect real fault and load conditions; and modeling more complex strip geometries beyond the simplified shapes used here.



Fusible Link Implementation in a Custom Cylindrical Cell Battery Pack

I used a coupled transient approach in Ansys Mechanical, combining electrothermal analysis for heat generation with thermal analysis for temperature rise. I defined nickel's resistivity as a temperature-dependent curve, so at each time step the solver computes Joule heating from the strip's current resistance, updates the temperature, then reads the new resistance off the curve before the next step.

This two-way coupling matters because nickel's resistivity climbs with temperature: as the strip heats, it generates more heat per amp, which accelerates the run toward the fusing point. Iterating over the simulation period captures the strip's full temperature-versus-time response, including this feedback behavior near failure.



Graph of Temperature Over Time

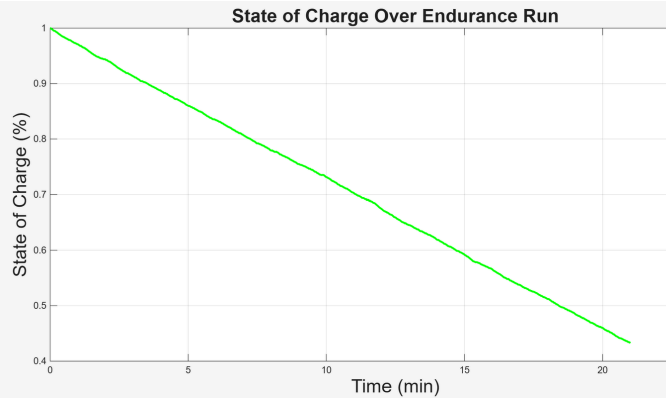
My Role: Full design and implementation (individual Project)
Software Used: Ansys Mechanical and Ansys Workbench

MATLAB simulation of tractive battery temperature for FSAE applications

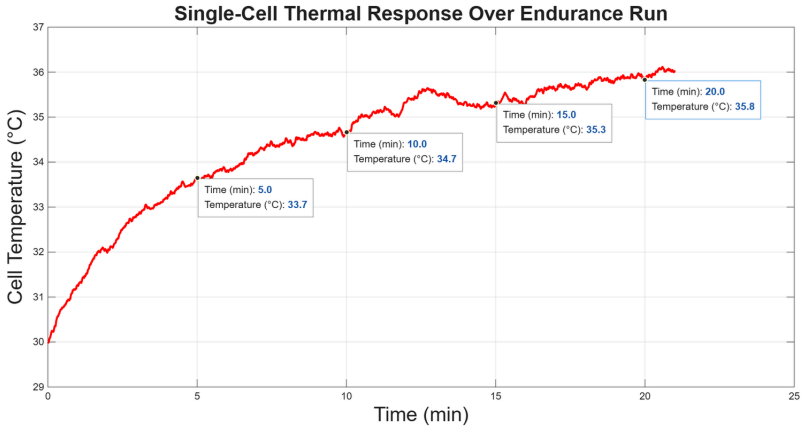
Objective

The objective of the simulation is to predict cell temperature and state of charge across a full endurance run for a theoretical Molicel P60C pack.

Using a real LapSim current profile and SOC-dependent DCIR from Molicel's published data, the model computes ohmic heat generation and temperature rise over time to determine whether the pack requires active cooling.



Graph of SOC over time per cell in a 20 minute endurance race

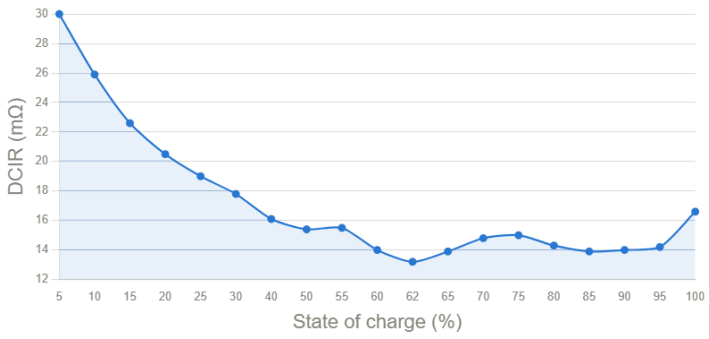


Graph of Temperature over time per cell in a 20 minute endurance race

Technical Details

The model solves a coupled two-state system (cell temperature and state of charge) using MATLAB. Heat generation is ohmic (I^2R), with per-cell current derived from a LapSim power trace divided across the 140S3P pack. DCIR is looked up as a function of SOC from Molicel's published P60C data, so resistance rises realistically as the pack depletes.

Heat rejection is modeled as convection to ambient, $Q_{loss} = hA \cdot (T - T_{amb})$, where hA is the lumped convective conductance intended as the calibration parameter against future bench data. SOC is tracked by coulomb counting ($dSOC/dt = -I/Q_{cell}$), which feeds back into the DCIR lookup and closes the Loop. The endurance run is built by repeating the LapSim lap profile across the full 20 laps for an endurance race.



DCIR vs SOC Curve for Molicel P60C, extracted from Molicel Website

Limitations

The model is quite limited. The cell is treated as a single lumped thermal mass; heat generation is ohmic only, omitting entropic heat; current is derived from power using a fixed nominal voltage, ignoring voltage sag; and DCIR is a function of SOC only, without temperature dependence. The convective term and material properties are typical Li-ion assumptions rather than measured values.

Its strength is how little it needs. From only a DCIR curve, it produces useful data before a physical cell exists, or is even released, making it a practical screening tool for comparing candidate cells ahead of any bench testing. Measuring DCIR and temperature rise on a real cell to calibrate hA and confirm the model is the next step.

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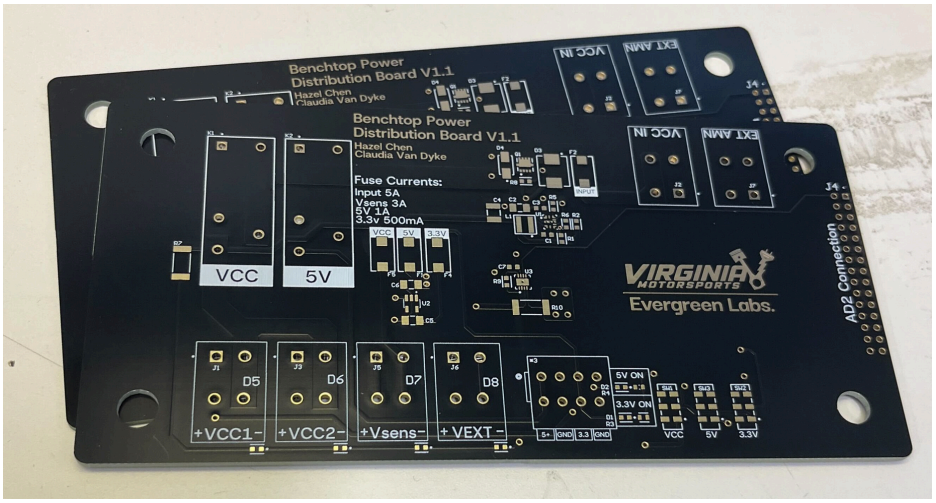
Benchtop Power Distribution Board

Objective

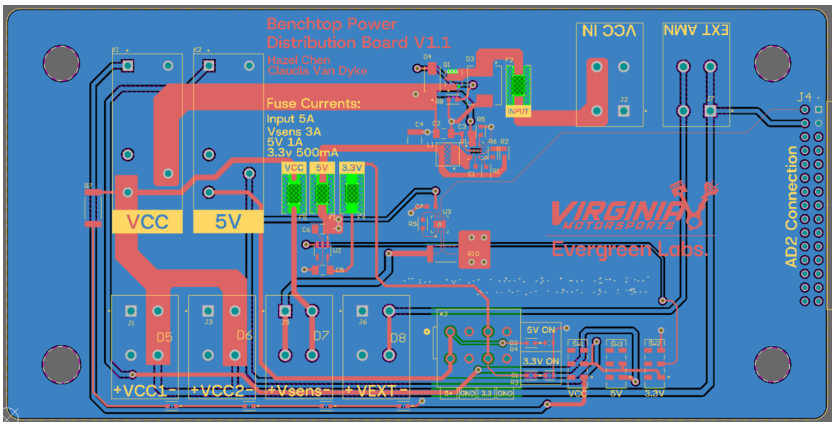
The objective of the Benchtop PDB is to facilitate power connections and testing of custom PCBs in a bench environment. It features individually toggleable power outputs, a 5V output, and a 3.3V output.

A TI LMR60420-Q1 buck converter steps the main input voltage (12V-36V) down to 5V, and an AP2112k LDO regulator then generates the 3.3V rail from this 5V supply.

The Benchtop PDB uses multiple WAGO board-to-wire quick-connect terminals, allowing a user to connect boards that require power easily.



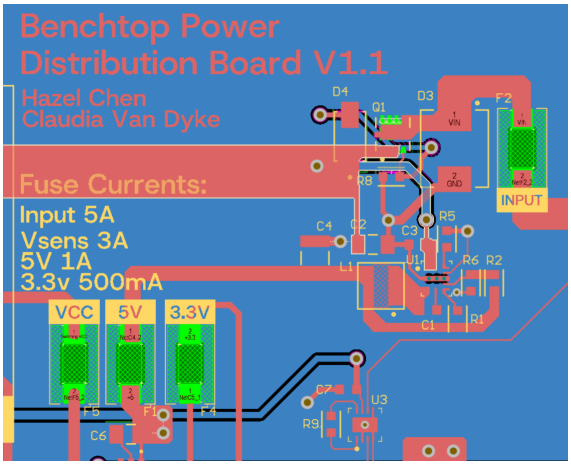
V1.1 PCBs



PCB Layout in Altium Designer

Technical Specifications

Input Voltage	12V-36V 5A max
Input Protection	TVS diode, MOSFET reverse polarity Protection, 5A fused
5V Output	5V, 1A max
3.3V Output	3.3V, 0.5A max
Current measuring	Shunt resistor and external ammeter support



Layout of Input Protection Circuitry

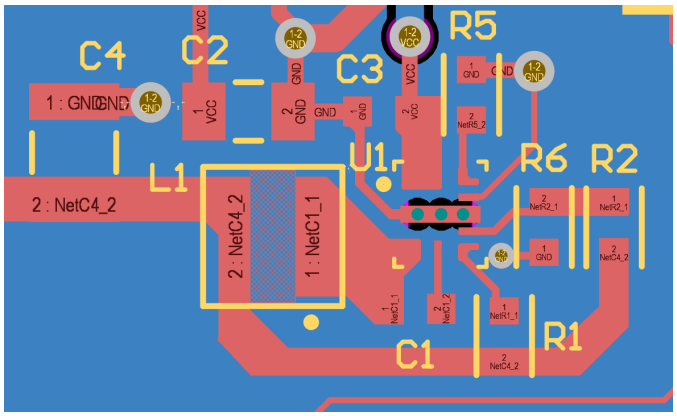
Power Protection

Each output channel is individually fused, and a 5A fuse also protects the main power input. The design includes reverse-polarity protection via a P-channel MOSFET (NTTFS5116PLTWG) that blocks current if reverse voltage is applied, avoiding the forward voltage drop penalty of a series diode solution. Transient-voltage protection is implemented with a TVS diode that clamps at 40V to safeguard connected hardware from input voltage spikes.

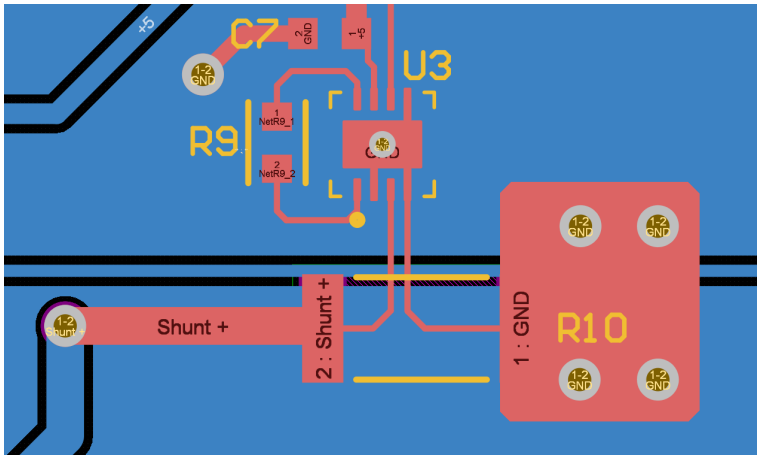
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Voltage Conversion

A TI LMR60420-Q1 buck converter steps the input voltage (12V–36V) down to a regulated 5V rail. An AP2112K LDO regulator then derives a 3.3V output from this 5V rail in a two-stage conversion topology, trading efficiency for improved noise performance on the 3.3V output. The 5V rail also powers two relays that independently switch the VCC and 5V outputs, allowing individual channels to be toggled without interrupting the remaining outputs.



TI LMR60420 Buck Converter Layout



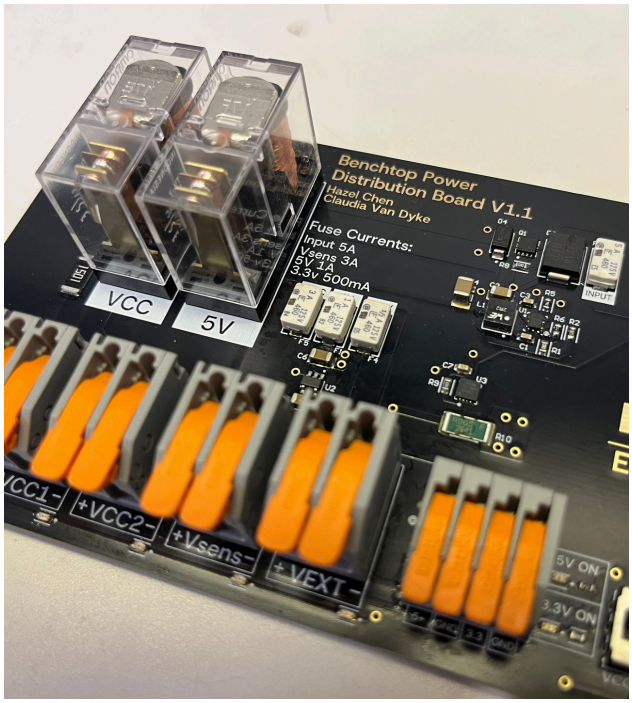
Shunt Resistor and Amplifier Layout

Assembly

PCBs and solder paste stencil were fabricated by JLCPCB. Solder paste was applied manually using a stencil, followed by individual pick-and-place of SMT components. Boards were reflowed in a reflow oven, then inspected and reworked as needed. Through-hole connections were hand-soldered to complete the assembly.

Current Sensing

The Benchtop Power Distribution Board includes two outputs that provide current sensing. The Vsens output uses an integrated shunt resistor and a connected Digilent Analog Discovery 2 to enable precise current measurement. The VEXT output allows the user to connect an external ammeter for quick and convenient measurements.



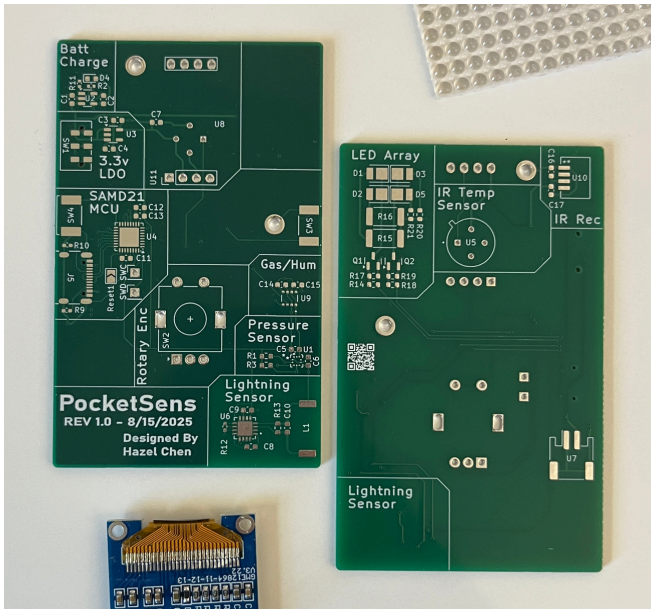
Fully Assembled PCB

PocketSens Portable Sensor Suite

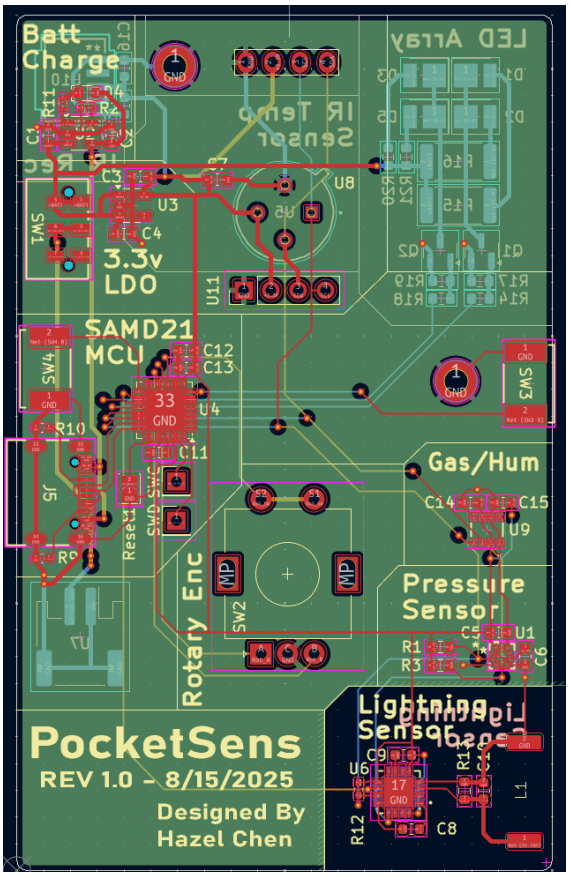
The objective of PocketSens is to provide users with a portable device that measures environmental and physical data beyond the capabilities of a smartphone, offering sensors for parameters such as pressure, temperature, humidity, gas, and lightning detection in one compact system.

Technical Specifications

MCU	ARM Cortex -M0+ SAMD21E18
Power and Supply	Single lithium-polymer cell, MCP73831 Charge IC, USBC input.
Gas/Humidity	BME 680, Gas (MOX), humidity ±3% RH, pressure ±1 hPa, temp ±1°C, I2C.
Lightning Detection	Lightning detection up to 40km, I2C, programmable filtering.
Infrared Sensor	TSOP6238 receiver, 0.4W transmitter, MLX90614 Non-Contact thermometer.
Altimeter	BMP390 Barometric Pressure and Altimeter (±0.25 m) sensor.



REV 1.0 Prototype PCB



PCB Layout in KiCad

Technical Details

PocketSens PCBs and stencil were fabricated by JLCPCB. Solder paste was applied manually using a stencil, followed by pick-and-place of SMT components. Boards were reflowed in a reflow oven, then inspected and reworked as needed. Through-hole components and certain SMT components were hand-soldered to complete the assembly.

J-Link EDU Mini was used to flash the bootloader, after which Arduino IDE could be used to upload code.

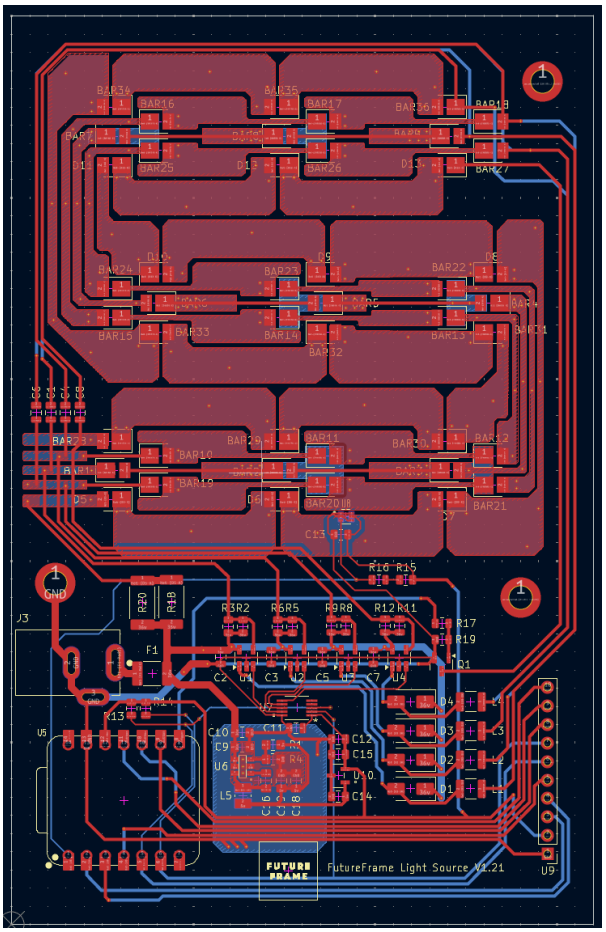
Future Frame LightCore V1.21

Objective

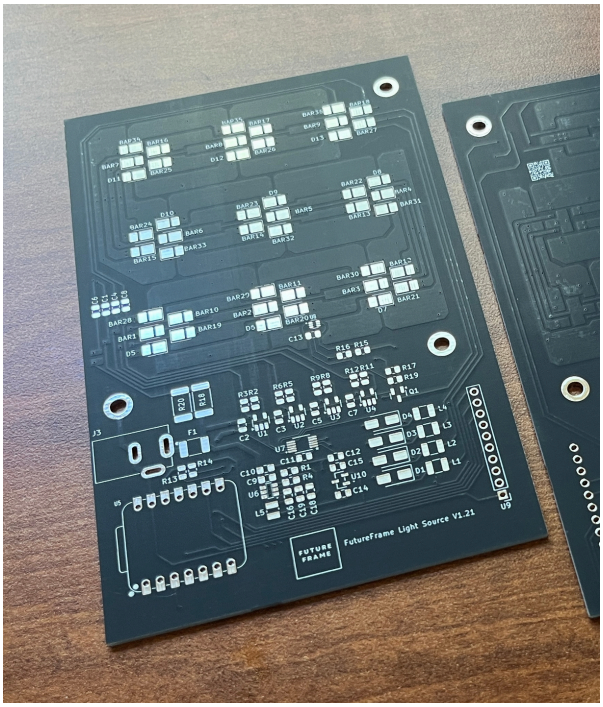
The objective of this project is to develop a precise and versatile light source for photographic film scanning. It uses five independent channels (RGB, white, and infrared) to deliver accurate color reproduction, uniform illumination, and added functions such as dust detection and IR transmission. With active temperature monitoring and efficient power control, the system supports high quality film digitization with greater flexibility.

Technical Specifications

Input Voltage	36V nominal
Voltage Conversion	AP64060 Buck Converter (36v-5v) AP7375 LDO (5v-3.3v)
Control	Seeed Xiao RP2040
LED Control	4x AL8860WT, DAC controlled for DC dimming.
Overtemp Protection	TMP302BDRLR, cutoff at 80°C
User Controls	0.96"OLED display, rotary encoder, mechanical switches
LEDs	3x RGB, White and infrared



V 1.21 Prototype PCB



V 1.21 Prototype PCB

A Seeed Studio XIAO microcontroller controls the LightCore on a custom PCB. Power is managed with a two-stage system that steps 36V down to 5V through a buck converter and then to 3.3V with an LDO. Four AL8860WT LED drivers and a four-channel DAC provide precise current control of the RGBW channels, while a MOSFET circuit drives the infrared channel for dust removal and IR transmission. A temperature sensor actively monitors temperature to maintain stable functionality. Together, these components form a reliable, high-precision platform for film scanning applications.

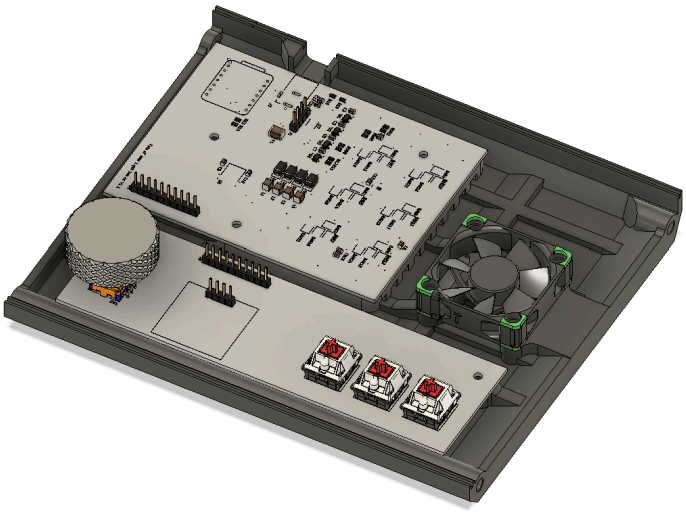
My Role: Full design and implementation (individual Project)
Software Used: KiCad Design Suite, Autodesk Fusion (CAD), Circuit Python

Future Frame LightCore V1.21 (Continued)

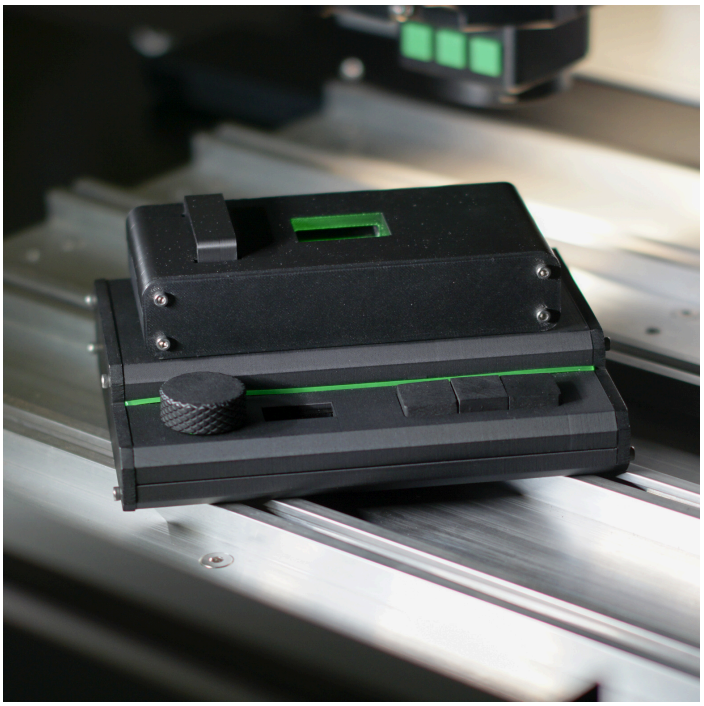
Mechanical Design

The light source is enclosed in a custom housing that combines a 3D printed body with sheet aluminum side walls for rigidity. Thermal management is supported by a waterjet aluminum heat dissipator mounted to the PCB. A two-stage diffusion system ensures even backlighting.

A secondary controller PCB is mounted within the housing and includes a rotary encoder, four switches, and an OLED display, providing direct user interaction and control of the light panel.



V1.2 CAD Design



Assembled V1.2 LightCore

Takeaways

This project helped me strengthen my skills in PCB design, SMT soldering, and 3D design for enclosures. I learned how to translate circuit requirements into a functional board layout, assemble and troubleshoot components during soldering, and design mechanical parts that fit and support the electronics.

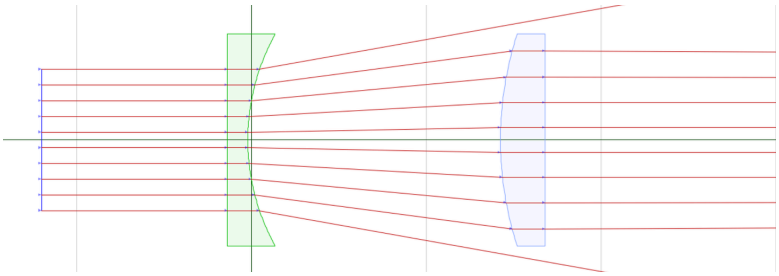
These experiences gave me a better understanding of how hardware and mechanical design work together in a complete system.

Future Frame Hybrid Viewfinder

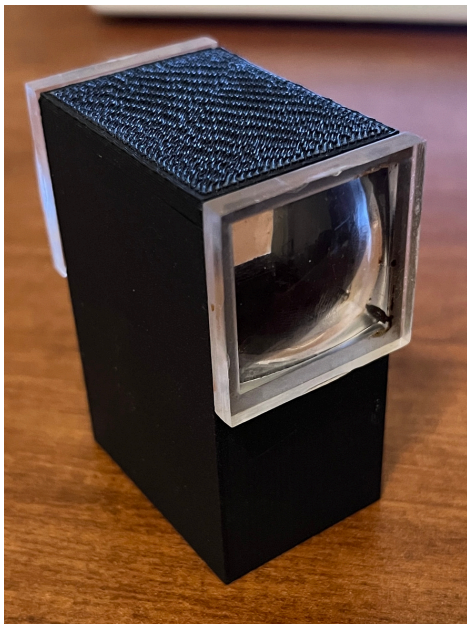
Objective

The objective of this project is to develop a hybrid viewfinder prototype that overlays an infinity-focused image in mid-air when looking through the viewfinder using custom-designed and manufactured optics.

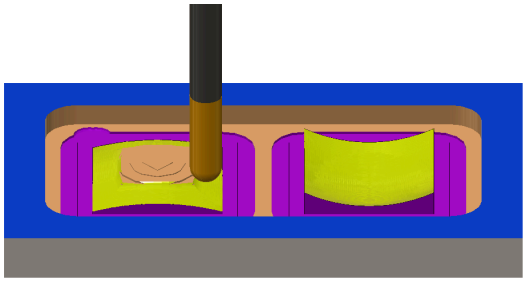
This design utilizes two acrylic lenses, one concave and one convex, that make up an afocal system. This arrangement is similar to a simple Galilean telescope, just reversed to achieve a demagnification effect



Simulation of the Convex-Concave Afocal System

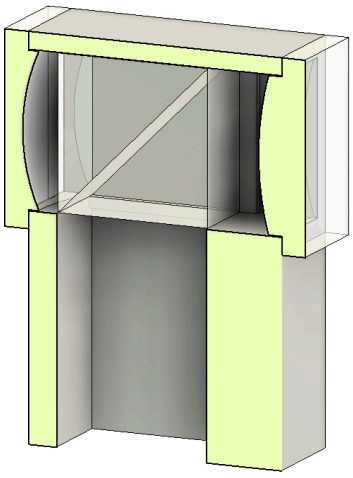


The acrylic lenses were custom-manufactured on a CNC router and polished to optical clarity

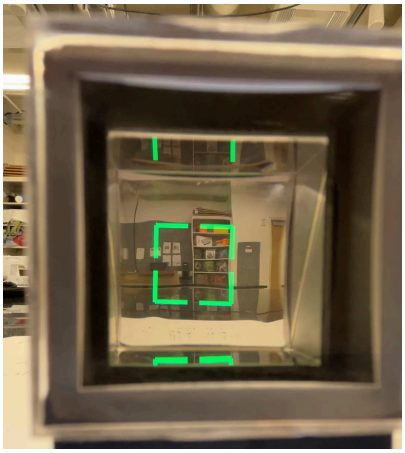


Fusion CAM Simulation

To achieve a HUD-like effect that focuses a display at infinity, a 45-degree beam splitter is used to project an image through the viewfinder. The design utilizes the rear lens element to focus the screen at infinity, which allows for everything in the viewfinder to be in focus at once.



CAD model Section View



Finished Prototype